SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

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QUESTIONS 18 / 18 completed

Sum(ODD,EVEN)

SUM(ODD,EVEN)

C Program to Find the Sum of Even and Odd Numbers.
Problem Description
The program takes the number N and finds the sum of odd and even numbers from 1 to N.

Testcases:
-5
10
4
100

PROGRAM EDITOR

C Run Save

```c
#include<stdio.h>
int main(){
  int  i,start,end;
  int evensum=0,oddsum=0;
  printf("Enter the start and end range: ");
  scanf("%d%d", &start, &end);
  for(i = start; i <= end;i ++){
    if(i %2==0)
      evensum = evensum+i;
    else
      oddsum = oddsum+i;
  }
  printf("Sum of Even numbers is:%d\n",evensum);
  printf("Sum of Odd numbers is:%d\n",oddsum);
}
```

OUTPUT

```
Enter the start and end range: Sum of Even numbers is:110
Sum of Odd numbers is:100
```

INPUT

1 20

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Natural

NATURAL

Write a program that prints the first n natural numbers using a for loop.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
int main(){
  int n,i;
  printf("Enter the number: ");
  scanf("%d",&n);
  printf("First %d natural numbers are:\n",n);
  for(i=1;i<=n;i++)
    printf("%d",i);
}
```

OUTPUT

```
Enter the number: First 10 natural numbers are:
12345678910
```

INPUT

10

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QUESTIONS 18 / 18 completed

Sum

SUM

Write a program that calculates the sum of the first n natural numbers using a while loop.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
int main(){
  int  n,i=1,sum=0;
  printf("Enter the value of n :");
  scanf("%d",&n);
  while(i<=n){
    sum = sum+i;
    i++;
  }
    printf("Sum of first %d natural numbers is=%d",n,sum);
}
```

OUTPUT

```
Enter the value of n :Sum of first 10 natural numbers is=55
```

INPUT

10

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Table

TABLE

Write a program that prints the multiplication table of a given number using a for loop.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
int main(){
  int i,n;
  printf("Enter the number: ");
  scanf("%d",&n);
  for(i=1;i<=10;i++){
    printf("%dX%d=%d\n",n,i,n*i);
  }
}
```

OUTPUT

```
Enter the number: 3X1=3
3X2=6
3X3=9
3X4=12
3X5=15
```

INPUT

3

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Prime

PRIME

Write a program that takes a number as input and determines whether it is a prime number or not using a while loop.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
int main() {
    long long n;
    long long i = 2;
    if (scanf("%lld", &n) != 1) return 0;
    if (n <= 1) {
        printf("Not Prime");
        return 0;
    }
    while (i * i <= n) {
        if (n % i == 0) {
            printf("Not Prime");
            return 0;
        }
        i++;
    }
    printf("Prime");
    return 0;
}
```

OUTPUT

```
Not Prime
```

INPUT

9

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QUESTIONS 18 / 18 completed

Count Vows

COUNT VOWS

Write a program that takes a string as input and counts the number of vowels in the string using a for loop.

PROGRAM EDITOR

C

Run Save

```
#include <stdio.h>
int main() {
    char str[1000];
    int i, count = 0;
    if (fgets(str, sizeof(str), stdin) == NULL) return 0;
    for (i = 0; str[i] != '\0'; i++) {
        char c = str[i];
        if (c >= 'A' && c <= 'Z') c = c - 'A' + 'a';
        if (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u') count++;
    }
    printf("%d", count);
    return 0;
}
```

OUTPUT

```
Security Error: Disallowed code pattern detected.
```

INPUT

ShaikMubarak

35°C
Partly sunny

Search

ENG
IN

74%

02:12:15 PM
06-08-2026

QUESTIONS 18 / 18 completed

Fibonacci

FIBONACCI

Write a program that prints the Fibonacci series up to a given number using a while loop.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main() {
    long long n, a = 0, b = 1, c;
    if (scanf("%lld", &n) != 1) return 0;
    while (a <= n) {
        printf("%lld", a);
        if (a + b <= n) printf(" ");
        c = a + b;
        a = b;
        b = c;
    }
    return 0;
}
```

OUTPUT

```
0 1 1 2 3 58
```

INPUT

```
8
```

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QUESTIONS 18 / 18 completed

Factorial

FACTORIAL

Write a program that takes a number as input and prints its factorial using a for loop.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
int main(){
  int i,n,fact=1;
  printf("Enter the number: ");
  scanf("%d",&n);
  for(i=1;i<=n;i++)
  fact = fact*i;
  printf("Factorial :%d\n",fact);
}
```

OUTPUT

```
Enter the number: Factorial :120
```

INPUT

5

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QUESTIONS 18 / 18 completed

Pyramid

PYRAMID

Write a program that prints a pyramid pattern of stars using nested loops.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
int main(){
  int i,j,n=5;
  for(i=1;i<=n;i++){
    for(j=1;j<=n-i;j++)
      printf(" ");
    for(j=1;j<=2*i-1;j++)
      printf("*");
    printf("\n");
  }
}
```

OUTPUT

```
    *
   ***
  *****
 *******
*********
```

INPUT

Your INPUT goes here!

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QUESTIONS 18 / 18 completed

Diamond

DIAMOND

Write a program that prints a diamond pattern of stars using nested loops.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
int main(){
  int i,j,n=5;
  for(i=1;i<=n;i++){
    for(j=1;j<=n-i;j++)
      printf(" ");
    for(j=1;j<=2*i-1;j++)
      printf("*");
    printf("\n");
  }
for(i=n-1;i>0;i--){
  for(j=1;j<=n-i;j++)
    printf(" ");
  for(j=1;j<=2*i-1;j++)
    printf("*");
  printf("\n");
}
}
```

OUTPUT

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

INPUT

Your INPUT goes here!

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C PROGRAMMING

80% − + Reset

QUESTIONS 18 / 18 completed

R- triangle

R- TRIANGLE

Write a program that prints a right-angled triangle pattern of stars using nested loops.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
int main(){
  int i,j,n=5;
  for(i=1;i<=n;i++){
    for(j=1;j<=i;j++)
      printf("*");
    printf("\n");
  }
}
```

OUTPUT

```
*
**
***
****
*****
```

INPUT

Your INPUT goes here!

QUESTIONS 18 / 18 completed

Hollow Square

HOLLOW SQUARE

Write a program that prints a hollow square pattern of stars using nested loops.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
int main(){
  int i,j,n=5;
  for(i=1;i<=n;i++){
    for(j=1;j<=n;j++){
      if(i==1||i==n||j==1||j==n)
        printf("*");
      else
        printf(" ");
  }
  printf("\n");
  }
  return 0;
}
```

OUTPUT

```
*****
*   *
*   *
*   *
*****
```

INPUT

Your INPUT goes here!

QUESTIONS 18 / 18 completed

Hollow Diamond

HOLLOW DIAMOND

Write a program that prints a hollow diamond pattern of stars using nested loops.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
int main()
{
    int n=5, i, j;
    scanf("%d", &n);
    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
            printf(" ");
        printf("*");
        if(i > 1)
        {
            for(j = 1; j <= 2 * i - 3; j++)
                printf(" ");
            printf("*");
        }
        printf("\n");
    }
    for(i = n - 1; i >= 1; i--)
    {
        for(j = 1; j <= n - i; j++)
```

OUTPUT

```
    *
   * *
  *   *
 *     *
*       *
 *     *
  *   *
   * *
    *
```

INPUT

Your INPUT goes here!

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QUESTIONS 18 / 18 completed

Pascal's triangle

PASCAL'S TRIANGLE

Write a program that prints a Pascal's triangle pattern using nested loops.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main() {
    int n=5, i, j;
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        for (j = 0; j < n - i - 1; j++) printf(" ");
        long long val = 1;
        for (j = 0; j <= i; j++) {
            printf("%lld", val);
            if (j < i) printf(" ");
            val = val * (i - j) / (j + 1);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
```

INPUT

Your INPUT goes here!

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QUESTIONS 18 / 18 completed

zigzag

ZIGZAG

Write a program that prints a zigzag pattern of stars using nested loops.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int total = 2 * n - 1;
    for (int i = 0; i < total; i++) {
        int stars;
        if (i < n) stars = i + 1;
        else stars = total - i;
        for (int j = 0; j < stars; j++) printf("*");
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
*
**
***
****
*****
****
***
**
*
```

INPUT

5

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QUESTIONS 18 / 18 completed

Spiral pattern

SPIRAL PATTERN

Write a program that prints a spiral pattern of numbers using nested loops.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
int main() {
    int n, i, j, top, bottom, left, right;
    int num = 1, total;
    if (scanf("%d", &n) != 1)
      return 0;
    total = n * n;
    top = 0;
    bottom = n - 1;
    left = 0;
    right = n - 1;
    while (num <= total) {
        for (i = left; i <= right && num <= total; i++)
            printf("%d ", num++);
        top++;
        if (top > bottom)
          break;
        for (i = top; i <= bottom && num <= total; i++)
            printf("%d ", num++);
        right--;
        if (left > right)
```

OUTPUT

```
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
```

INPUT

10

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QUESTIONS 18 / 18 completed

Armstrong

ARMSTRONG

Write a program to find the given number is Armstrong or Not.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
int main(){
  int num,temp,digit,sum=0;
  printf("Enter the number: ");
  scanf("%d",&num);
  temp = num;
  while(temp>0){
    digit = temp%10;
    sum+=digit*digit*digit;
    temp/=10;
  }
  if(sum == num)
    printf("Armstrong Number");
  else
    printf("Not an Armstrong Number");
}
```

OUTPUT

```
Enter the number: Armstrong Number
```

INPUT

153

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QUESTIONS 18 / 18 completed

Perfect

PERFECT

Write a program to find the given number is Perfect number or Not.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
int main() {
    int n, i;
    int sum = 0;
    if (scanf("%d", &n) != 1)
      return 0;
    if (n <= 0) {
        printf("Not Perfect Number");
    }
    for (i = 1; i <= n / 2; i++) {
        if (n % i == 0) sum += i;
    }
    if (sum == n)
        printf("Perfect Number");
    else
        printf("Not Perfect Number");
    return 0;
}
```

INPUT

24

OUTPUT

```
Not Perfect Number
```